LIBBY — MANUSCRIPTS INVENTORY

Manuscripts considered — aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2

Master inventory: every paper reviewed in this case

Generated 2026-07-10

Manuscripts considered — aml-mds-related-rr-tp53-aberrant-hla-p

Master inventory: 17 clinical (14 included, 3 considered & excluded) + 16 pre-clinical (14 included, 2 considered & excluded) + 16 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Dumbrava/Schram (2026) — New England Journal of Medicine

Reference:PMID 41740031 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586, selective p53 Y220C reactivator)
- **Indication / model:** Advanced/metastatic solid tumors harbouring TP53 Y220C (2L+); Heavily pretreated locally advanced/metastatic solid tumors with a TP53 Y220C mutation; PYNNACLE phase 1 dose escalation/optimization. Solid tumors only — no myeloid patients; responses restricted to KRAS wild-type tumors.
- **Design:** Phase 1 single-group dose-escalation/optimization, monotherapy
- **n:** 77
- **Effect size:** 20 % — Overall response (CR/PR)
- **Toxicities:** nausea (n=77, 58%); vomiting (n=77, 44%); blood creatinine increased (n=77, 39%); fatigue (n=77, 39%); anaemia (n=77, 36%); anaemia G>=3 (n=77, 16%); treatment-related AE discontinuation (2/77, 3%)
- **Case fit:** cross_tumor_only
- **Notes:** Proof-of-concept for Y220C-selective p53 reactivation, but all data are in solid tumors and this patient's TP53 status is unconfirmed (2019 mutant vs 2026 negative on a limited assay) with Y220C not established. A low-positive/unconfirmed biomarker is a weaker predictor than a tier-matched confirmed one; the myeloid trial NCT06616636 has no results. Double contingency: confirm TP53 aberration, then confirm it is Y220C.

Gyurkocza/Giralt (2025) — Journal of Clinical Oncology

Reference:PMID 39298738 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Iomab-B (131I-apamistamab, anti-CD45 radioimmunotherapy conditioning)
- **Indication / model:** Active relapsed/refractory AML, age >=55, planned allogeneic HCT (2L+); Patients >=55 with active R/R AML randomised to a 131I-apamistamab-led regimen before alloHCT vs conventional care (with crossover to apamistamab allowed on non-response). Enrolled patients heading to a first alloHCT.
- **Design:** Randomized open-label phase 3 (SIERRA), with crossover
- **n:** 153
- **Effect size:** 17.1% vs 0% % — Durable complete remission (>=180 days)
- **Variance:** p=<0.0001
- **Toxicities:** any treatment-related adverse event G>=3 (n=76, 59.7%); any treatment-related adverse event G>=3 (n=77, 59.2%)
- **Case fit:** partial
- **Notes:** Highest-grade evidence that CD45-targeted RIT can carry active high-burden R/R AML into an allo-HCT, directly relevant to conditioning a second transplant with an 85%-blast marrow. SIERRA enrolled patients heading to a first transplant, so prior-HCT eligibility is the axis to verify. Per-term AE breakdown not in the abstract.

Issa/Stein (2025) — Journal of Clinical Oncology

Reference:PMID 39121437 · **Type:** clinical · **Inclusion:** excluded — Biomarker mismatch and out of feature scope: revumenib acts through the menin-KMT2A interaction and its approval / AUGMENT-101 data are in KMT2A-rearranged (or NPM1-mutant) acute leukemia. This patient has KMT2A amplification (4 copies), a distinct lesion for which menin-inhibitor benefit is unproven, and no menin/KMT2A-rearrangement target is listed among the patient's targetable_features.

- **Intervention:** Revumenib (SNDX-5613, menin inhibitor)
- **Notes:** Excluded: Biomarker mismatch and out of feature scope: revumenib acts through the menin-KMT2A interaction and its approval / AUGMENT-101 data are in KMT2A-rearranged (or NPM1-mutant) acute leukemia. This patient has KMT2A amplification (4 copies), a distinct lesion for which menin-inhibitor benefit is unproven, and no menin/KMT2A-rearrangement target is listed among the patient's targetable_features. — Reviewed at the user's explicit request for an honest menin-inhibitor read. AUGMENT-101 is a real, positive trial in KMT2A-rearranged disease; the exclusion is on eligibility/biomarker grounds, not evidence quality.

—/— (2025)

Reference: — · **Type:** clinical · **Inclusion:** excluded — Same biomarker mismatch as revumenib and out of feature scope: ziftomenib is approved in NPM1-mutant AML and studied in KMT2A-rearranged disease. This patient has neither a documented NPM1 mutation nor a KMT2A rearrangement (the lesion is KMT2A amplification), so no proven ziftomenib biomarker applies and none is among the targetable_features.

- **Intervention:** Ziftomenib (KO-539, menin inhibitor)
- **Notes:** Excluded: Same biomarker mismatch as revumenib and out of feature scope: ziftomenib is approved in NPM1-mutant AML and studied in KMT2A-rearranged disease. This patient has neither a documented NPM1 mutation nor a KMT2A rearrangement (the lesion is KMT2A amplification), so no proven ziftomenib biomarker applies and none is among the targetable_features. — Logged for completeness of the menin-inhibitor picture the profile asked about. No dedicated publication cited here to avoid an unverifiable identifier; COMET-001 enrolls NPM1-mutant/KMT2A-rearranged disease.

—/— (2025) — Blood (ASH 2025 trial-in-progress abstract)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** CBX-250 (TCR-mimetic bispecific T-cell engager targeting CG1/HLA-A*02:01)
- **Indication / model:** Relapsed/refractory AML, high-risk MDS, and CMML (2L+); HLA-A*02:01-positive R/R myeloid leukemia; prior allogeneic HCT permitted once at least 60 days have elapsed.
- **Design:** First-in-human open-label phase 1 dose-escalation, single-agent (CROSSCHECK-001, NCT06994676)
- **Effect size:** — — Trial in progress; no efficacy readout reported
- **Toxicities:** Trial in progress, so there is no clinical safety or efficacy readout yet. As a CD3-engaging T-cell engager the anticipated class toxicity is cytokine release syndrome; the preclinical program reported high CG1/Cathepsin-G antigen selectivity with relative sparing of normal myeloid cells.
- **Case fit:** partial
- **Notes:** On the patient's primary HLA-A*02:01 axis, but gated on the same pending HLA typing as the rest of that axis. Distinct advantage over the TCR-T options: an off-the-shelf T-cell engager with no autologous or donor cell manufacturing. Added after the initial screen, which missed the trial because CG1/Cathepsin-G was not a nominated feature. Identifiers left null: the ASH 2025 trial-in-progress abstract DOI could not be verified

through available channels during this run (fail-closed per the reference-check protocol), so no identifier was guessed. — (primary endpoint not extracted)

Carter/Andreeff (2025) — Blood

Reference:PMID 40608889 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586, Y220C-selective p53 reactivator)
- **Indication / model:** TP53-Y220C AML cell lines and primary AML samples
- **Design:** Rezatapopt binds the surface pocket created by the Y220C substitution and refolds mutant p53 into a wild-type conformation that reactivates p53 target genes.
- **Effect size:** moderate — In Y220C AML, rezatapopt restored wild-type p53 conformation and transcriptional output but drove little apoptosis on its own, because induced MDM2 and XPO1 dampen p53 activity and the reactivated protein does not engage BCL-2. Adding venetoclax (or MDM2/XPO1 inhibition) brought back strong apoptotic killing.
- **Variance:** vs Rezatapopt alone vs rezatapopt plus venetoclax (or MDM2/XPO1 inhibition); translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Tumor- and variant-matched (Y220C AML), but two contingencies stay open here: confirm the TP53 aberration, then confirm it is Y220C. An unconfirmed biomarker is a weaker predictor than a tier-matched confirmed one. The data predict rezatapopt in myeloid disease needs a venetoclax or MDM2/XPO1 partner rather than single-agent use.

Puzio-Kuter/Poyurovsky (2025) — Cancer Discovery

Reference:PMID 39945593 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586) discovery and characterization
- **Indication / model:** Biochemical/structural assays with solid-tumor cell-line and xenograft models
- **Design:** Rezatapopt and related compounds occupy the Y220C surface pocket and refold the mutant, restoring DNA binding and the downstream p53 transcriptional program.
- **Effect size:** moderate — Structural, transcriptomic, and functional data confirmed Y220C conformational correction and recovery of the wild-type p53 program, with antitumor activity as a single agent and combined with immunotherapy. The work characterizes the molecule now in a registrational solid-tumor trial.
- **Variance:** translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Characterization sits largely in solid-tumor models; the myeloid-specific behavior (limited single-agent apoptosis) is covered by Carter 2025. Company-authored (PMV).

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** FH-WT1-E50 (anti-WT1 TCR/CD8ab T-cells) + azacitidine
- **Indication / model:** AML with measurable residual disease (post-therapy) (2L+); HLA-A*02:01, WT1-expressing
- **Design:** Phase 1, autologous HLA-A*02:01-restricted anti-WT1 TCR-T plus azacitidine
- **Effect size:** — — Safety, WT1-directed activity

- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** 131I-apamistamab RIT conditioning + allo-HCT
- **Indication / model:** Acute leukemia, radioimmunotherapy conditioning for allo-HCT (2L+); CD45 (targeted conditioning)
- **Design:** Phase 2/3, 131I-apamistamab conditioning + fludarabine/cyclophosphamide/TBI
- **Effect size:** — — Engraftment, remission, survival
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CBX-250 (TCR-mimetic bispecific T-cell engager targeting CG1/HLA-A*02:01)
- **Indication / model:** R/R AML / high-risk MDS / CMML / CML (CROSSCHECK-001, first-in-human) (2L+); HLA-A *02:01 required; target is CG1 (Cathepsin-G-derived peptide) presented on HLA-A*02:01
- **Design:** Open-label phase 1 first-in-human dose-escalation, single-agent, subcutaneous
- **Effect size:** — — Safety, tolerability, MTD/RP2D, preliminary clinical activity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Krakow/Bleakley (2024) — Blood

Reference:PMID 38683966 · **Type:** clinical · **Inclusion:** included

- **Intervention:** HA-1-specific CD8+/CD4+ TCR-T (Bleakley platform)
- **Indication / model:** Persistent/recurrent AML, ALL, MDS or JMML after allogeneic HCT (2L+); HLA-A*02:01-positive recipients with recipient HA-1(H)-positive / donor HA-1-negative mismatch; disease recurrence after allo-HCT; donor-derived product after lymphodepletion.
- **Design:** Single-arm phase 1, donor-derived HA-1 TCR-T post-HCT
- **n:** 9
- **Effect size:** 4/9 CR patients — CR after lymphodepletion + HA-1 TCR-T
- **Toxicities:** dose-limiting toxicity (0/9)
- **Case fit:** partial
- **Notes:** Closest published precedent for the patient's primary handle and the same mechanism as TSC-100 (NCT05473910). Feasibility/safety phase 1, not powered for efficacy. Per-term safety table not retrievable: publisher/PMC WebFetch blocked, PMC OA efetch returned front-matter only, and NCT03326921 has no results posted on ClinicalTrials.gov.

Daver/Kantarjian (2024) — Lancet Oncology

Reference:PMID 38423051 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pivekimab sunirine (IMGN632, CD123 antibody-drug conjugate)
- **Indication / model:** Relapsed/refractory acute myeloid leukaemia, CD123-positive (2L+); CD123+ R/R AML, ECOG 0-1, one or more prior AML treatments; recommended phase 2 dose expansion n=29.
- **Design:** First-in-human phase 1/2 dose-escalation/expansion, monotherapy
- **n:** 91
- **Effect size:** 21 % — Overall response rate at recommended phase 2 dose
- **Variance:** 95% CI 8–40
- **Toxicities:** febrile neutropenia G>=3 (3/29, 10%); infusion-related reaction G>=3 (2/29, 7%); anaemia G>=3 (2/29, 7%); veno-occlusive disease (2/68); treatment-related death G5 (1/68, 1%)
- **Case fit:** partial
- **Notes:** Single-agent CD123 ADC activity in R/R AML that matches the patient's CD123 positivity; quantitative antigen-density selection is advisory and still pending. VOD signal is relevant given a planned second transplant.

Lee/Wiederschain (2024) — Blood

Reference:doi:10.1182/blood-2024-201699 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CBX-250 (TCR-mimetic bispecific T-cell engager targeting CG1/HLA-A*02:01)
- **Indication / model:** AML cell lines + aggressive primary-AML PDX (FLT3-ITD/DNMT3A/NPM1) mouse model
- **Design:** TCR-mimetic bispecific antibody binding the CG1 nonamer (FLLPTGAEA, from Cathepsin G) presented on HLA-A*02:01, cross-linked to CD3 to redirect polyclonal T cells against blasts. CG1 sits high on AML/myeloid-leukemia cells and low or absent on normal myeloid cells, and the mechanism does not rely on graft-versus-leukemia, so it works off-the-shelf independent of a donor product.
- **Effect size:** strong — Sub-nanomolar EC50 T-cell-mediated killing of leukemia lines spanning a range of antigen density, with no T-cell activation or cytotoxicity against healthy-donor neutrophils, CD34+ cells, or PBMCs. In the primary-AML PDX, CBX-250 cut tumor burden and extended survival over control, active down to 0.005 mg/kg.
- **Variance:** vs vehicle/untreated control arm in the PDX model; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** ASH 2024 conference abstract (Blood 144 Suppl 1:208), not a peer-reviewed full paper; no PMID and per-arm n not reported. Antigen and HLA-A*02:01 restriction line up with the patient's AML and A*02:01 axis, but that allele is still pending typing and no clinical activity has been shown.

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BSB-1001 (HA-1-directed TCR-T)
- **Indication / model:** Recurrent AML / ALL / MDS undergoing HLA-matched allo-HCT (2L+); HLA-A*02:01 + HA-1 recipient-positive / donor-negative
- **Design:** Phase 1/2 dose-escalation and expansion of a donor-derived HA-1 TCR-T after allo-HCT
- **Effect size:** — — Safety, dose, anti-leukemic activity
- **Case fit:** partial

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tagraxofusp + cladribine + cytarabine
- **Indication / model:** CD123-positive relapsed/refractory AML (2L+); CD123+
- **Design:** Phase 1/2, tagraxofusp plus low-intensity cladribine/cytarabine
- **Effect size:** — — Safety, response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586) + azacitidine
- **Indication / model:** TP53 Y220C-mutant MDS / AML (2L+); TP53 Y220C
- **Design:** Phase 1b, rezatapopt plus azacitidine
- **Effect size:** — — Safety, response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Garcia-Manero/Sallman (2023) — Lancet Haematology

Reference:PMID 36990622 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Eprenetapopt (APR-246, pan-p53 reactivator)
- **Indication / model:** TP53-mutated acute myeloid leukaemia, treatment-naive (1L); Treatment-naive TP53-mutated AML, ECOG 0-2; eprenetapopt + venetoclax +/- azacitidine. Frontline population, not R/R — a line mismatch for this patient.
- **Design:** Phase 1 dose-finding and expansion, single-arm
- **n:** 49
- **Effect size:** 64 % — Overall response (eprenetapopt + venetoclax + azacitidine)
- **Variance:** 95% CI 47–79
- **Toxicities:** febrile neutropenia G>=3 (23/49, 47%); thrombocytopenia G>=3 (18/49, 37%); leukopenia G>=3 (12/49, 25%); anaemia G>=3 (11/49, 22%); treatment-related death G5 (1/49, 2%)
- **Case fit:** weak
- **Notes:** Pan-p53 reactivator, not Y220C-selective, and studied frontline rather than in R/R disease. The profile flags it as low-yield: the eprenetapopt+azacitidine phase 3 in TP53-mutant MDS missed its complete-remission primary endpoint and myeloid development has largely stopped. Kept so the board can weigh the pan-reactivator track against Y220C-selective rezatapopt.

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Pivekimab sunirine + FLAG-Ida (fludarabine, cytarabine, idarubicin, G-CSF)
- **Indication / model:** AML / MPAL / MDS (incl. R/R) (2L+); CD123+
- **Design:** Phase 1, pivekimab sunirine added to FLAG-Ida
- **Effect size:** — — Safety, response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** clinical · **Inclusion:** excluded — Population mismatch and program discontinuation: the CD123xCD3 bispecific was studied in MRD-positive AML/MDS and terminated. The MRD setting does not match this patient's overt high-burden relapse (~85% marrow blasts), and no pivotal efficacy publication anchors an evidence row.

- **Intervention:** Vibecotamab (XmAb14045, CD123xCD3 bispecific)
- **Notes:** Excluded: Population mismatch and program discontinuation: the CD123xCD3 bispecific was studied in MRD-positive AML/MDS and terminated. The MRD setting does not match this patient's overt high-burden relapse (~85% marrow blasts), and no pivotal efficacy publication anchors an evidence row. — On-mechanism for the CD123 handle but no usable clinical-evidence publication; the CD123 bispecific evidence base is carried by the flotetuzumab row.

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TSC-100 (donor-derived HA-1-specific TCR-T)
- **Indication / model:** R/R AML / MDS / ALL undergoing HLA-matched allo-HCT (ALLOHA) (2L+); HLA-A*02:01 + HA-1(H) recipient-positive / donor-negative
- **Design:** Open-label phase 1, donor-derived TCR-T given after HLA-matched allo-HCT, vs standard-of-care control arm
- **Effect size:** — — Safety, donor chimerism, relapse
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TSC-101 (donor-derived HA-2-specific TCR-T)
- **Indication / model:** R/R AML / MDS / ALL undergoing HLA-matched allo-HCT (ALLOHA) (2L+); HLA-A*02:01 + HA-2 recipient-positive / donor-negative
- **Design:** Open-label phase 1, donor-derived TCR-T given after HLA-matched allo-HCT, vs standard-of-care control arm
- **Effect size:** — — Safety, donor chimerism, relapse

- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Vibecotamab (XmAb14045, CD123xCD3 bispecific)
- **Indication / model:** MRD-positive AML / MDS (2L+); CD123xCD3 (no biomarker selection)
- **Design:** Phase 2, vibecotamab in MRD-positive disease (terminated)
- **Effect size:** — — MRD conversion
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Uy/DiPersio (2021) — Blood

Reference:PMID 32929488 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Flotetuzumab (MGD006, CD123xCD3 DART bispecific)
- **Indication / model:** Primary induction failure / early-relapse refractory AML (2L+); Adults with R/R AML; 42 in dose-finding and 46 at the 500 ng/kg/day RP2D. Efficacy reported for the 30 primary-induction-failure/early-relapse patients treated at RP2D.
- **Design:** Multicenter open-label phase 1/2 single-arm
- **n:** 88
- **Effect size:** 26.7 % — CR/CRh in PIF/early-relapse patients at RP2D
- **Toxicities:** CRS / infusion-related reaction
- **Case fit:** partial
- **Notes:** CD123xCD3 bispecific with salvage activity in chemo-refractory AML, close to this patient's state; program has since been deprioritised by MacroGenics, so it is mechanism precedent rather than an open route. Per-grade CRS breakdown is in the full text, not the abstract.

—/— (2021)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NTLA-5001 (WT1-directed TCR-T)
- **Indication / model:** Acute myeloid leukemia (2L+); HLA-A*02:01, WT1+
- **Design:** Phase 1/2 CRISPR-engineered WT1 TCR-T (terminated)
- **Effect size:** — — Safety, dose (program halted)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586) +/- pembrolizumab
- **Indication / model:** Advanced solid tumors, TP53 Y220C (2L+); TP53 Y220C
- **Design:** Phase 1/2 (PYNNACLE), rezatapopt monotherapy and with pembrolizumab
- **Effect size:** — — Objective response, safety
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Chapuis/Greenberg (2019) — Nature Medicine

Reference:PMID 31235963 · **Type:** clinical · **Inclusion:** included

- **Intervention:** WT1-TCRc4 EBV-specific donor CD8+ T cells (TTCR-C4)
- **Indication / model:** High-risk AML, relapse prophylaxis after allogeneic HCT (maintenance); HLA-A*02:01-positive, WT1-expressing high-risk AML; TCR inserted into EBV-specific donor CD8+ T cells and infused prophylactically post-HCT (not for active disease).
- **Design:** Phase 1/2 single-arm with concurrent comparative cohort
- **n:** 12
- **Effect size:** 100 % — Relapse-free survival vs comparator cohort
- **Toxicities:** Prophylactic product engineered into EBV-specific CD8+ T cells to minimise GVHD; abstract reports long-term persistence and polyfunctionality without a per-term AE breakdown.
- **Case fit:** weak
- **Notes:** Relapse-prevention (prophylaxis) evidence, not active-relapse salvage; gated on HLA-A*02:01 and superseded operationally by FH-WT1-E50. Comparator is a historical/concurrent cohort, so the survival contrast is confounded. Per-term safety not in the abstract; PMC/publisher full text not retrievable via available channels.

Pemmaraju/Konopleva (2019) — New England Journal of Medicine

Reference:PMID 31018069 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Tagraxofusp (SL-401, CD123-directed IL-3/diphtheria-toxin fusion)
- **Indication / model:** Blastic plasmacytoid dendritic-cell neoplasm (BPDCN), untreated or relapsed (1L); CD123-overexpressing BPDCN; 32 first-line and 15 previously treated; registration cohort for FDA approval. Not the patient's disease (AML) — mechanism/off-label precedent for the CD123 handle.
- **Design:** Open-label multicohort phase 2 single-arm
- **n:** 47
- **Effect size:** 72 % — Combined CR + clinical CR (first-line, 12 μg/kg)
- **Toxicities:** capillary leak syndrome (n=47, 19%); ALT increased (n=47, 64%); AST increased (n=47, 60%); hypoalbuminemia (n=47, 55%); peripheral edema (n=47, 51%); thrombocytopenia (n=47, 49%)
- **Case fit:** cross_tumor_only
- **Notes:** Registration study behind tagraxofusp's BPDCN approval; carries the CD123 mechanism as off-label precedent, not on-label AML support. Rates are all-grade from the abstract; per-grade table is in the full text.

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CD123-CD33 cCAR T cells

- **Indication / model:** Relapsed/refractory high-risk AML / MDS (2L+); CD123+ and/or CD33+
- **Design:** Early phase 1 compound CAR-T co-targeting CD123 and CD33
- **Effect size:** — — Safety, dose
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CLL-1/CD33/CD123-directed CAR-T
- **Indication / model:** Relapsed/refractory AML (2L+); CLL-1 and/or CD33 and/or CD123
- **Design:** Phase 1/2 multi-target CAR-T against CLL-1, CD33, CD123
- **Effect size:** — — Safety, response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Ziftomenib (menin inhibitor)
- **Indication / model:** Relapsed/refractory AML (NPM1-mutant or KMT2A-rearranged) (2L+); NPM1-mutant / KMT2A rearrangement
- **Design:** Phase 1/2 first-in-human (COMET-001), ziftomenib monotherapy
- **Effect size:** — — Response, safety
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Dossa/Bleakley (2018) — Blood

Reference:PMID 29051183 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** HA-1-specific CD8+/CD4+ TCR-T (Bleakley construct)
- **Indication / model:** Human primary leukemia co-culture with lentivirally engineered CD8+ and CD4+ TCR-T
- **Design:** Lentiviral HA-1 TCR transgene bundled with a CD8 coreceptor to arm CD4+ cells, an inducible caspase-9 off-switch, and a CD34/CD20 epitope for selection and tracking; only memory T cells are used, to limit native-TCR alloreactivity.
- **Effect size:** moderate — Bleakley's group built the four-part HA-1 TCR construct that became the clinical product, and both CD4+ and CD8+ transduced T cells responded to primary HA-1-positive leukemia. The design couples leukemia-directed cytotoxicity with a safety switch and a selectable marker.
- **Variance:** translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Construct-development paper; the reported responses are in-vitro against primary leukemia, without an

in-vivo efficacy arm. Same eligibility gate as TSC-100: HLA-A*02:01 plus recipient HA-1(H)-positive / donor-negative, both unresolved here.

Kovtun/Chittenden (2018) — Blood Advances

Reference:PMID 29661755 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Pivekimab sunirine (IMGN632, CD123 antibody-drug conjugate)
- **Indication / model:** AML cell lines, patient-derived AML samples, and multiple AML xenograft models
- **Design:** Humanized anti-CD123 antibody G4723A conjugated to a DNA mono-alkylating indolinobenzodiazepine (IGN) payload; CD123 binding drives internalization and payload delivery to blasts and stem cells.
- **Effect size:** strong — IMGN632 killed AML lines and patient samples at low picomolar potency regardless of multidrug-resistance or disease status, and stayed active at doses well under those affecting normal myeloid progenitors. It controlled disease across several AML xenografts, with a wider therapeutic window than a crosslinking-payload comparator.
- **Variance:** vs Crosslinking-IGN-payload ADC (X-ADC) and normal myeloid progenitors; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Company-authored (ImmunoGen) preclinical package. The mono-alkylating payload was picked for the normal-progenitor sparing that sets the clinical therapeutic window; antigen-density selection stays advisory pending quantitative CD123.

Petrov/Ma (2018) — Leukemia

Reference:PMID 29515236 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CD123-CD33 compound CAR-T (dual-antigen; iCell cCAR, NCT04156256)
- **Indication / model:** Four human AML xenograft mouse models; CD33/CD123 compound CAR-T
- **Design:** One T cell expressing two discrete scFv CARs, one against CD123 and one against CD33, so escape of either antigen still leaves a functional receptor; an alemtuzumab-binding domain allows the cells to be eliminated.
- **n:** 4 AML xenograft models
- **Effect size:** strong — The compound CAR-T killed each antigen-positive population and gave better survival across four AML mouse models than single-antigen targeting. Covering CD123 (LSC-associated) alongside CD33 was framed to blunt antigen-escape relapse.
- **Variance:** vs Single-antigen CAR pressure / untreated mice; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Xenograft-only, company-originated (iCell) construct with no post-allo-HCT data; CAR-T after a prior transplant carries its own eligibility and CRS considerations. Directly the antigen-escape hedge the profile raises given clonal evolution at this patient's relapse.

Tawara/Shiku (2017) — Blood

Reference:PMID 28860210 · **Type:** clinical · **Inclusion:** included

- **Intervention:** WT1-siTCR gene-transduced lymphocytes (TBI-1301)

- **Indication / model:** Refractory AML and high-risk MDS (2L+); HLA-A*24:02-restricted WT1-specific TCR (retroviral vector with siRNA silencing of endogenous TCR); two dose-escalation cohorts, transfers 4 weeks apart followed by WT1 peptide vaccine.
- **Design:** First-in-human phase 1, dose-escalation
- **n:** 8
- **Effect size:** 0 normal-tissue AEs — Safety and T-cell persistence
- **Toxicities:** on-target off-tumor normal-tissue toxicity (0/8)
- **Case fit:** weak
- **Notes:** A*24:02 fallback lever, relevant only if the patient is A*02:01-negative. Small early-phase Japanese cohort; efficacy signals are transient blast reductions rather than remissions.

—/— (2016)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BPX-701 (PRAME TCR-T with iCasp9 safety switch) + rimiducid
- **Indication / model:** Acute myeloid leukemia / MDS (2L+); HLA-A*02:01, PRAME+
- **Design:** Phase 1/2 rimiducid-controllable PRAME TCR-T (terminated)
- **Effect size:** — — Safety, dose (program halted)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Frolova/Konopleva (2014) — British Journal of Haematology

Reference:PMID 24942980 · **Type:** preclinical · **Inclusion:** excluded — Model system is advanced-phase chronic myeloid leukemia and its stem cells, not AML. Testa 2005 carries the tumor-matched, IL-3R-density-dependent mechanism for this AML case, so the CML paper adds nothing on the CD123 axis here.

- **Intervention:** SL-401 / SL-501 (CD123-directed) in chronic myeloid leukemia
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Model system is advanced-phase chronic myeloid leukemia and its stem cells, not AML. Testa 2005 carries the tumor-matched, IL-3R-density-dependent mechanism for this AML case, so the CML paper adds nothing on the CD123 axis here. — On-mechanism for the CD123 handle but tumor-mismatched; the tagraxofusp/DT-IL3 preclinical row is carried by Testa 2005 in primary AML blasts.

Christopeit/Schmid (2013) — Journal of Clinical Oncology

Reference:PMID 23918951 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Second allogeneic HCT for relapse after first allo-HCT
- **Indication / model:** Hematologic relapse of acute leukemia after first allogeneic HCT (2L+); EBMT registry of 179 second transplants for relapse after a first transplant from matched related (n=75) or unrelated (n=104) donors; analysis of donor change. The patient's first HCT was from a matched sibling, so the related-donor subgroup is the relevant reference.
- **Design:** Retrospective registry cohort
- **n:** 179
- **Effect size:** 25 % — 2-year overall survival after HSCT2

- **Toxicities:** Registry outcome study; per-term toxicity and non-relapse mortality are not broken out in the abstract. Remission duration after HSCT1 and disease stage at HSCT2 were the dominant survival drivers.
- **Case fit:** partial
- **Notes:** Anchors realistic HSCT2 expectations for this matched-sibling patient (~37% 2-year OS in the related-donor subgroup), and shows donor change is a valid but not clearly superior option. Long remission after HSCT1 (this patient ~6 years) favours survival; active high-burden disease at HSCT2 works against it.

Ochi/Yasukawa (2011) — Blood

Reference:PMID 21673345 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** WT1-siTCR gene-transduced lymphocytes (TBI-1301 basis)
- **Indication / model:** Patient autologous leukemia lysis assays and human AML xenograft in mice
- **Design:** Retroviral WT1-TCR vector carrying siRNAs that silence the endogenous TCR chains, which reduces mispairing and raises surface expression of the introduced WT1-specific receptor.
- **Effect size:** moderate — The siTCR cells outperformed a conventional WT1 TCR and lysed patients' autologous leukemia while sparing normal hematopoietic progenitors. Transferred cells controlled leukemia in a xenograft without impairing human hematopoiesis.
- **Variance:** vs Conventional WT1-TCR (WT1-coTCR) without endogenous-TCR silencing; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** This is the HLA-A*24:02-restricted WT1 siTCR behind TBI-1301, so it only becomes relevant if the patient is A*02:01-negative and A*24:02-positive, neither established. Xenograft plus patient-cell data, no syngeneic model.

Jin/Lock (2009) — Cell Stem Cell

Reference:PMID 19570512 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Anti-CD123 antibody targeting of AML leukemic stem cells (CD123-LSC rationale)
- **Indication / model:** NOD/SCID xenograft of primary human AML
- **Design:** A neutralizing anti-CD123 antibody (7G3) blocks IL-3 signaling on CD34+CD38- AML cells, impairs their homing to marrow, and engages mouse innate effectors against engrafted stem cells.
- **Effect size:** moderate — 7G3 cut AML-LSC engraftment, lowered burden in mice with established leukemia, and impaired secondary transplantation, so the stem-cell compartment itself was reached. Early in-vivo validation that antibody targeting of CD123 hits AML LSCs.
- **Variance:** translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The antibody is 7G3 (the CSL362/talacotuzumab lineage), not pivekimab's G4723A, so it supports CD123-as-LSC-target rather than the specific ADC. Naked-antibody mechanism; no payload contribution tested.

Lambert/Bykov (2009) — Cancer Cell

Reference:PMID 19411067 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Eprenetapopt (APR-246 / PRIMA-1) pan-p53 reactivator
- **Indication / model:** Recombinant mutant p53 proteins and human tumor cell lines in vitro

- **Design:** PRIMA-1/APR-246 converts to methylene quinuclidinone (MQ), which covalently binds cysteine thiols in the p53 core domain and shifts mutant p53 toward a wild-type-like, DNA-binding fold.
- **Effect size:** moderate — Lambert traced reactivation to covalent thiol adducts formed by the MQ metabolite, active across several different p53 mutants rather than one pocket. This thiol-based, sequence-nonselective mechanism is what separates eprenetapopt from a Y220C-pocket binder.
- **Variance:** translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Because the mechanism is covalent and not variant-selective, benefit does not concentrate on Y220C; the profile flags eprenetapopt as low-yield after the phase 3 TP53-MDS miss. In vitro only. Kept to contrast the pan-reactivator route against variant-selective rezatapopt.

Boeckler/Fersht (2008) — Proc Natl Acad Sci U S A

Reference:PMID 18650397 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Y220C-pocket stabilizer PhiKan083 (structural basis for Y220C reactivators)
- **Indication / model:** In-silico screen plus X-ray crystallography and thermal-stability assays of the Y220C p53 mutant
- **Design:** The Y220C mutation opens a surface crevice that destabilizes the p53 DNA-binding domain; a small molecule (PhiKan083) docks into that crevice and raises the mutant's melting temperature, slowing denaturation.
- **Effect size:** weak — Fersht's group found a drug-like compound that binds the Y220C-specific pocket and thermodynamically stabilizes the mutant protein. This was the first structural proof that the crevice is druggable, the premise every Y220C-selective reactivator now builds on.
- **Variance:** translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Biophysics and structure with an early tool compound (PhiKan083), not rezatapopt and not in any tumor model. It grounds why a Y220C-selective corrector is mechanistically separate from a pan-reactivator.

Schmid/Rocha (2007) — Journal of Clinical Oncology

Reference:PMID 17909197 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Donor lymphocyte infusion (DLI) for AML relapse after allo-HCT
- **Indication / model:** AML in first haematological relapse after allogeneic HCT (2L+); EBMT registry of 399 patients with AML in first relapse post-HCT; 171 treated with DLI vs 228 without. Multivariate analysis of survival drivers among DLI recipients.
- **Design:** Retrospective comparative registry cohort
- **n:** 399
- **Effect size:** 21% vs 9% % — 2-year overall survival, DLI vs no DLI
- **Variance:** p=0.04
- **Toxicities:** GVHD is the characteristic DLI toxicity; the abstract reports survival rather than per-term AE rates. Clinical benefit was limited to a minority — best when DLI was given in remission or with favourable karyotype, worst in aplasia or active disease.
- **Case fit:** partial
- **Notes:** Sets honest expectations for sibling DLI in this patient: relapse >5 months after HCT (here ~6 years) favours benefit, but 85% marrow blasts and adverse cytogenetics put him in the active-disease/high-burden

group where 2-year OS is ~15%. Argues for cytoreduction before any DLI rather than DLI alone.

Schmid/Kolb (2005) — Journal of Clinical Oncology

Reference:PMID 16110027 · **Type:** clinical · **Inclusion:** included

- **Intervention:** FLAMSA-RIC sequential chemotherapy + reduced-intensity allo-HCT + prophylactic DLI
- **Indication / model:** High-risk AML and MDS (mostly progressive/refractory disease) (2L+); 75 consecutive high-risk patients (59 progressive/refractory), 49% unfavourable karyotype; 31 with HLA-identical sibling donor. Fludarabine/cytarabine/amsacrine cytoreduction, then 4 Gy TBI/ATG/cyclophosphamide RIC, then prophylactic DLI from day +120.
- **Design:** Prospective single-arm cohort
- **n:** 75
- **Effect size:** 88 % — Complete remission after the sequential regimen
- **Toxicities:** GVHD and prophylactic-DLI-associated GVHD are the principal risks; survival was best with high CD34+ cell dose and limited GVHD. Per-term toxicity is not tabulated in the abstract.
- **Case fit:** partial
- **Notes:** The FLAMSA-RIC sequencing this patient's plan is built on: it takes refractory AML with adverse cytogenetics straight to a reduced-intensity allo-HCT and reports outcomes for that hardest subgroup equal to lower-risk patients. 31 of 75 used HLA-identical sibling donors, matching the patient's donor situation.

Xue/Stauss (2005) — Blood

Reference:PMID 16020516 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** HLA-A*02:01-restricted WT1 TCR-T (A*02:01 WT1 TCR class)
- **Indication / model:** NOD/SCID xenograft engrafted with human leukemia; WT1-TCR gene-transduced human T cells
- **Design:** Retroviral transfer of an HLA-A*02:01-restricted WT1-specific TCR into human T cells, redirecting them against WT1 peptide presented on leukemia.
- **Effect size:** moderate — WT1-TCR-transduced human T cells cleared human leukemia in NOD/SCID mice while leaving normal CD34+ progenitors intact. An early in-vivo signal that an A*02:01 WT1 TCR can carry antileukemic activity.
- **Variance:** translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** A distinct A*02:01 WT1 TCR from the Stauss lab, not the Greenberg C4 receptor used in the WT1-TCRc4 clinical product, so it supports the class rather than the specific construct. Xenograft only; gated on A*02:01, unresolved.

Testa/Frankel (2005) — Blood

Reference:PMID 15928038 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Tagraxofusp / DT-IL3 class (CD123-directed diphtheria-toxin fusion)
- **Indication / model:** Primary AML blasts from 34 patients, ex vivo cytotoxicity
- **Design:** DT388 (the catalytic and translocation domains of diphtheria toxin) fused to IL-3 enters blasts through IL-3Rα (CD123), so intracellular toxin delivery and killing scale with receptor density; a K116W IL-3 variant

raises binding affinity.

- **n:** n=34 patient AML samples
- **Effect size:** moderate — Across 34 primary AML samples, killing by the DT-IL3 fusions tracked IL-3Rα/β expression, and the higher-affinity K116W variant was more potent per mole. Receptor level worked as a simple biochemical predictor of sensitivity.
- **Variance:** vs DT388IL-3 wild-type vs higher-affinity DT388IL-3[K116W] variant; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Ex-vivo primary-blast assay, no in-vivo arm; the fusion is the DT-IL3 class that tagraxofusp belongs to. Density-dependent killing is the mechanistic reason the profile asks for quantitative CD123 before committing.

Hamann/Bernstein (2002) — Bioconjugate Chemistry

Reference:PMID 11792178 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Gemtuzumab ozogamicin (anti-CD33 calicheamicin conjugate)
- **Indication / model:** CD33+ human myeloid leukemia cell lines and human tumor xenografts
- **Design:** Humanized anti-CD33 antibody (hP67.6) conjugated to N-acetyl-gamma-calicheamicin through an acid-hydrolyzable linker; CD33 binding drives internalization, lysosomal payload release, and calicheamicin-induced DNA double-strand breaks.
- **Effect size:** strong — The conjugate was selectively toxic to CD33-expressing leukemia lines and active against xenografts, with potency that followed CD33 antigen. It set out the antibody-calicheamicin design that became Mylotarg.
- **Variance:** translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Conjugate-design paper; CD33-dependent uptake means antigen density and drug-efflux status both modulate response, which is why quantitative CD33 is requested. The calicheamicin payload carries hepatic/VOD risk into the peri-transplant setting.

Bykov/Selivanova (2002) — Nature Medicine

Reference:PMID 11875500 · **Type:** preclinical · **Inclusion:** excluded — Original PRIMA-1 discovery; the covalent MQ reactivation mechanism it opened is resolved for this dossier by Lambert 2009, which pins the molecular basis. Redundant for the pan-reactivator contrast and superseded by that mechanism paper.

- **Intervention:** PRIMA-1 (original discovery of the APR-246 scaffold)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Original PRIMA-1 discovery; the covalent MQ reactivation mechanism it opened is resolved for this dossier by Lambert 2009, which pins the molecular basis. Redundant for the pan-reactivator contrast and superseded by that mechanism paper. — Foundational compound-discovery paper, not variant-selective; kept out to avoid duplicating the eprenetapopt mechanism already carried by Lambert 2009.

Sievers/Mylotarg Study Group (2001) — Journal of Clinical Oncology

Reference:PMID 11432892 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Gemtuzumab ozogamicin (Mylotarg, CD33 ADC)

- **Indication / model:** CD33-positive AML in untreated first relapse (2L+); CD33+ AML in first relapse, no antecedent haematologic disorder, median age 61; three pooled open-label single-arm trials. Population is less heavily pretreated than this patient (no prior FLAG-IDA/venetoclax failure or extramedullary disease).
- **Design:** Pooled phase 2 single-arm (three open-label trials)
- **n:** 142
- **Effect size:** 30 % — Overall remission (CR + CRp)
- **Toxicities:** hyperbilirubinemia G>=3 (n=142, 23%); hepatic transaminase elevation G>=3 (n=142, 17%); infection G>=3 (n=142, 28%); mucositis G>=3 (n=142, 4%); severe nausea/vomiting G>=3 (n=142, 11%)
- **Case fit:** partial
- **Notes:** On-label, on-mechanism CD33 option, but single-agent depth in first relapse is modest (~30%) and this patient is further along with FLAG-IDA/venetoclax failure and extramedullary disease. Hepatic/VOD signal is relevant peri-transplant; quantitative CD33 density still pending.

den Haan/Goulmy (1998) — Science

Reference:PMID 9461441 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** HA-1-directed minor-histocompatibility-antigen TCR-T (Bleakley/TSC-100 platform)
- **Indication / model:** Human cell-line expression profiling and minor-H-antigen-specific CTL recognition assays
- **Design:** HA-1 is an HLA-A*02:01-restricted minor histocompatibility antigen encoded by a diallelic gene; the immunogenic allele differs from its counterpart by a single amino acid, and the peptide is presented mainly on cells of hematopoietic origin.
- **Effect size:** null — The paper mapped HA-1 to a diallelic gene whose disparity reduces to one amino acid, and located expression on hematopoietic-lineage cells rather than epithelium. A donor T-cell response against recipient HA-1(H) can therefore reach leukemia while largely sparing the tissues that drive GVHD.
- **Variance:** translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Antigen-discovery work, not a therapeutic model. The graft-versus-leukemia-without-GVHD benefit hinges on a recipient HA-1(H)-positive / donor-negative mismatch plus HLA-A*02:01, neither genotyped in this pair yet; HA-2 (myosin-derived) is the parallel mHAg behind TSC-101.